INCH-POUND

MIL-A-85097/3C(OS)
5 December 1994
SUPERCEDING
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17 October 1985

MILITARY ASSOCIATED DETAIL SPECIFICATION

AIRCREW ESCAPE PROPULSION SYSTEMS (AEPS) ROCKET MOTOR (LOADED ASSEMBLY) MARK 79 MOD 1

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Mark 79 Mod 1 Rocket Motor described herein shall consist of this specification and the latest issue of MIL-A-85097.

1. SCOPE

1.1 Scope. This document covers requirements unique to the Rocket Motor Mark 79 Mod 1 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents are in addition to those listed in the basic specification, MIL-A-85097, and form a part of this specification to the extent specified herein.

SPECIFICATIONS

NAVAL SURFACE WARFARE CENTER (CAGE Code 14083)

511-174TD-004 Propellant, NOSIH-BC-11

DRAWINGS

NAVAL AIR SYSTEMS COMMAND (CAGE Code 30003)

672AS200 Rocket Motor Assembly Mk 79 Mod 1

672AS231 Motor Subassembly

(Application for copies of specifications and drawings should be addressed to the Commander, Indian Head Division, Naval Surface Warfare Center, Attn: Code 8410P, 101 Strauss Avenue, Indian Head, MD 20640-5035.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Code 8420, 101 Strauss Avenue, Indian Head, MD 20640-5035 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

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3. REQUIREMENTS

3.1 General. The requirements in this specification, unless in direct conflict with the basic specification, shall be considered additional, not substitute. The Mark 79 Mod 1 Rocket Motor shall conform to Drawing 672AS200. If there is a direct conflict with the cover specification, this slash sheet shall take precedence.

3.2 Preproduction.

3.2.1 Propellant aging. When tested in accordance with MIL-A-85097, the propellant shall meet the aging requirements specified in 511-174TD-004.

3.3 Preproduction/production.

3.3.1 Propellant. The propellant shall conform to 511-174TD-004. The contractor shall submit all test results in accordance with 511-174TD-004 and MIL-A-85097.

3.3.2 Ballistic performance. The ballistic performance shall conform to table I.

TABLE I. Ballistic performance requirements.

Characteristic	Value		Temperature (±5°F)
	Min	Max	
Total impulse (lb-sec)	580	640	-40, 160
Action time (ms)	190	320	-40, 160
Thrust (lb)	-	3800	-40, 160
Ignition delay (ms)	-	25	-40, 160
Initiation pressure (psig)	400	650	-40, 160
Time to shear (ms)	-	10	-40, 160
Rise time (ms) (see 6.2.1)	-	30	-40, 160

4. QUALITY ASSURANCE PROVISIONS

4.1 Preproduction/production.

4.1.1 Conditioning temperatures. The conditioning temperatures and tolerance shall be as follows:

a. Cold: $-40 \pm 5°F$
b. Hot: $160 \pm 5°F$

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4.1.2 Conditioning period. Time for conditioning prior to static firing or during temperature cycling shall be as follows:

a. Static firing: Not less than 4 hours
b. Temperature cycling: Not less than 5 hours

4.1.3 Radiographic inspection. Radiographic inspection shall be as follows:

a. Inspect the propellant grain assembly at the head and aft end. Four longitudinal views shall be taken at each end, 45° apart, as specified by drawing 672AS231.

b. Inspect the rocket motor assembly in two views 90°apart, as specified by drawing 672AS200.

5. PACKAGING

(See latest revision of MIL-A-85097 and the applicable contract for packaging and marking requirements.)

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The MK 79 Mod 1 Seatback Rocket Motor assists in ejecting personnel from the T/AV-8B and A-7 aircraft during an emergency.

6.2 Definition.

6.2.1 Rise time. The interval from 10% to 75% of the maximum thrust during the initial rise on the thrust curve occurring just after ignition.

6.3 Changes from previous issue. Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing Activity:
Navy - OS
(Project 1377-NE84)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-A-85097/3 C	2. DOCUMENT DATE (YYMMDD) 5 December 1994

3. DOCUMENT TITLE
AIRCREW ESCAPE PROPULSION SYSTEMS (AEPS) ROCKET MOTOR (LOADED ASSEMBLY) MARK 79 MOD 1

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
C. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) DSN (*If applicable*)	7. DATE SUBMITTED (*YYMMDD*)

8. PREPARING ACTIVITY

a. NAME Commander, Indian Head Division Naval Surface Warfare Center	b. TELEPHONE (*Include Area Code*) (1) Commercial 301-744-4700 (2) DSN 354-4700
c. ADDRESS (*Include Zip Code*) Standardization Team (Code 840M) 101 Strauss Avenue Indian Head, MD 20640-5035	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888

DD FORM 1426, FEB 1999 (EG) PREVIOUS EDITION IS OBSOLETE.

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